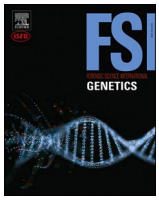
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Inter-laboratory evaluation of the VISAGE enhanced tool and models for age estimation from blood and buccal cells

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A B S T R A C T

Over the past decade, numerous assays for forensic age estimation based on the analysis of DNA methylation markers have been developed, demonstrating significant potential for use in criminal investigations. Despite these advancements, only few comprehensive evaluation studies were published. In this study, we present findings of an extensive inter-laboratory evaluation of the VISAGE Enhanced Tool and its associated statistical models for epigenetic age estimation in blood and buccal swabs. Six laboratories conducted reproducibility, concordance, and sensitivity assessments using DNA methylation controls alongside blood and saliva samples to evaluate the tool's technical performance. Results demonstrated consistent and reliable DNA methylation quantification across all participating laboratories, with the tool maintaining sensitivity even with a DNA input of 5 ng for bisulfite conversion. To evaluate the age estimation models, 160 blood and 100 buccal swab samples were analysed in three laboratories. The models achieved mean absolute errors (MAEs) of 3.95 years for blood and 4.41 years buccal swabs, which represents an increase of ~0.7 years for both tissues to the results from the original VISAGE testing set. When comparing results of each laboratory with the original VISAGE testing set, significant differences were found only for age estimation results from blood of one laboratory with an under-estimation of chronological age observed within the entire range tested at that laboratory. When excluding this laboratory, the MAE decreased to 3.1 years (N = 89). No significant differences among laboratories were found for buccal swabs. Overall, this study confirms that the VISAGE Enhanced Tool performs robust DNA methylation

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quantification and reliable age prediction, however protocol and model validation within each laboratory is required upon implementation.

1. Introduction

Estimation of the chronological age of an unknown individual has a long tradition in forensic investigations, with its origins in the evaluation of morphological characteristics of unidentified human remains [1]. In the past decade, various molecular markers have been proposed to improve age prediction accuracy and allow age estimation from biological material from traces [2]. Currently, the most accurate results from DNA samples are achieved through the quantification of DNA methylation (DNAm) levels at age correlated CpG sites [3–5]. Forensic age estimation assays, which target a small set of DNAm age markers (typically < 20) can predict the chronological age with mean absolute errors (MAEs) of approximately three to five years, depending on the analysed sample type [6–10]. This approach enables age prediction from minimal material, such as a bloodstain, thereby broadening the applicability of forensic age estimation in criminal investigations.

With the growing interest of the forensic community in DNAm based age estimation, various assays have been developed. These assays focus on different DNAm markers, DNAm detection methods and statistical models for age prediction [2]. Notably, DNAm patterns are cell-type specific and most of the developed models are applicable only to certain tissue types [11]. Additionally, the different assays (e.g., based on minisequencing [7,12,13], pyrosequencing [14–16] or massively parallel sequencing (MPS) [6,17,18]) applied to quantify the DNAm levels might show method-specific biases [19,20], which makes it difficult to transfer DNAm results between different assays.

Consequently, most forensic laboratories perform DNAm based age estimation using assays developed in-house and only few validation studies have been published so far [21–23]. However, small changes in the laboratory workflow can affect the final DNAm quantification, as shown by a collaborative study involving 12 laboratories to assess a SNaPshot assay for age estimation. The study showed that the Genetic Analyzer instrument used for minisequencing, the employed PCR buffers as well as the volume of bisulfite converted DNA can affect the final DNAm quantification results [21]. Additionally, GEDNAP proficiency tests showed systematic as well as marker-dependent differences between laboratories for minisequencing and pyrosequencing approaches [22,23]. When implementing a published method into the laboratory workflow, such factors may affect the accuracy of the age prediction model leading to increased MAEs [24]. These findings demonstrate the need for inter-laboratory validation of proposed methods and extended model testing to allow for the implementation of an assay into routine forensic casework.

The VISAGE Enhanced Tool for age estimation from somatic cells (herein, VISAGE Enhanced Tool) allows for the simultaneous analysis of eight well validated age markers (*KLF14*, *TRIM59*, *MIR29B2CHG*, *FHL2*, *ELOVL2*, *EDARADD*, *PDE4C* and *ASPA*) by targeted bisulfite MPS. The assay was developed by the VISAGE consortium (VISible Attributes through GEnomics, https://www.visage-h2020.eu/) and comprises age prediction models for blood, buccal cells and bones, which showed MAEs of 3.2, 3.7 and 3.4 years in the initial tests, respectively [25]. Furthermore, age estimation with the VISAGE Enhanced Tool is not significantly affected by heavy alcohol consumption [26] and, more recently, an age estimation model for cartilage samples was introduced, expanding the applicability of the VISAGE Enhanced Tool on another forensically relevant sample type [27]. Additionally, it was demonstrated that the eight VISAGE markers are suitable as tissue prediction markers, successfully differentiating between blood, buccal swab, bone and cartilage samples [27].

This study presents the collaborative efforts of the VISAGE consortium comprising six academic and police laboratories to evaluate the VISAGE Enhanced Tool. The laboratories participated in two collaborative exercises (Phase 1 and 2), which mainly focused on the evaluation of the technical aspects of the tool using DNAm controls and blood and saliva samples. Furthermore, the VISAGE age tools were tested for implementation in three governmental laboratories within the INFER project ("Introduction of forensic genomic tools for estimating Appearance, Ancestry and Age"). Whereas saliva was used to mimic casework samples in the technical evaluation, this project focused on the analysis of 160 blood and 100 buccal cell samples to test the associated predictive models for their performance across three laboratories.

2. Materials and methods

2.1. Study design of the VISAGE collaborative exercises

The collaborative exercises conducted within the VISAGE project were performed in two consecutive phases, which differed in the number of participating laboratories. Phase 1 and 2 aimed at the developmental validation of the MPS assay (Phase 1; N = 5 laboratories), followed by further implementation tests using a higher number of blood and saliva samples (Phase 2; N = 6 laboratories). Each laboratory received a detailed protocol to avoid additional variability in DNAm quantification due to different kits or buffers used and carried out one MiSeq FGx run per validation phase. An overview of the performed experiments and samples included in each study is provided below and can be found in Table 1. Duplicates prepared in this study refer to sample-replicates processed simultaneously, beginning with the bisulfite conversion step that followed DNA extraction and quantification.

2.2. DNAm control samples

Artificially methylated DNA standards (DNAm controls) were used to assess DNAm quantification for linearity, reproducibility and sensitivity. These DNAm control samples were prepared using the Human Methylated & Non-Methylated (WGA) DNA Set (Zymo Research, Irvine, CA, USA), diluted to 20 ng/µl in low TE. The two DNA preparations were then combined at different volume ratios to create standards with nominal DNAm percentage (**DNAm%**) levels of 0 %, 5 %, 10 %, 25 %, 50 %, 75 % and 100 %, respectively. These DNAm standards were prepared at the organizing laboratory and sent to participating laboratories.

2.3. Concordance and mock casework samples

Blood and saliva samples used for concordance and mock casework studies (for information on age and sex see Supplementary Table S1) were purchased from Lee BioSolutions (Maryland Heights, MO, USA), distributed through Hölzel Diagnostika Handels GmbH (Cologne, Germany). Blood sample A and B in Phase 1 were purchased from Biotrend (Cologne, Germany). DNA for first concordance evaluations (Phase 1), were extracted from 200 µl blood or saliva using the QIAamp DNA Mini Kit (Qiagen, Hilden, Germany), eluted in 200 µl elution buffer and quantified with the Qubit dsDNA HS Assay Kit on a Qubit 2.0 Fluorometer (both Thermo Fisher Scientific, Waltham, MA, USA, hereafter referred to as TFS). Stains serving as mock casework samples were prepared from the same samples by pipetting 100 µl blood or saliva on Whatman FTA cards (Merck, Darmstadt, Germany). Samples for Phase 1 were processed in duplicates by two partner laboratories and as single reactions by the organizing institute (Table 1). For Phase 2, cotton swabs (Prionics forensix, Voigtländer GmbH; Blumenberg, Germany) of 100 µl blood or saliva were prepared and each laboratory received five blood

and five saliva samples which were analysed in duplicates. DNA extraction and quantification from stains and swabs was performed by participating laboratories following their established protocols.

2.4. Samples used for VISAGE age model evaluation

Two sample sets collected separately were used for age model evaluation. The VISAGE-evaluation set consisted of 49 EDTA whole venous blood samples and 47 buccal swabs (Whatman Omniswabs), which were collected under the study number 1086/2017. All buccal swabs were collected from the same donors as blood samples. The study was approved by the ethics commission of the Medical University of Innsbruck and all volunteers provided written informed consent. DNA extraction of blood samples was performed with the QIAamp DNA Mini Kit (Qiagen) using the spin protocol for DNA purification from blood and an elution volume of 200 µl. DNA extraction of buccal swabs was performed with the same kit and an adapted spin protocol for buccal swabs with an extended incubation time at 56°C of 30 min and an elution volume of 25 µl. Blood samples were quantified by real-time PCR [28]. DNA quantification of buccal swabs was performed using the Qubit dsDNA HS kit (TFS). The INFER sample set consisted of 111 blood and 53 buccal cell samples, collected from 111 healthy volunteers aged 18–80 years. Sample collection was conducted in accordance with an ethics document approved by the Internal Security Fund (ISF) and the study was approved by the ethics commission of the Ulm University (study number 137/19). All samples were collected with written informed consent from the donors. Blood samples consisted of venous EDTA blood, which were stored frozen until extraction. Buccal swabs were stored dried until extraction. DNA extraction from buccal swabs was performed using the Maxwell® "FFS Nucleic Acid Extraction Kit" on the Maxwell® 16 system, and the Maxwell® "RSC PureFood GMO and Authentication Kit" on the Maxwell® 48 system (all Promega, Walldorf, Germany), following the manufacturer's recommendations. Blood samples were extracted using the QIAamp DNA Blood Midi Kit (Qiagen). DNA quantification was conducted using the Quantifiler Trio Kit on a 7500 real-time PCR System instrument (both TFS), according to the manufacturer's guidelines. All INFER samples were processed in duplicates.

2.5. Bisulfite conversion and PCR

Bisulfite conversions were performed using the Premium Bisulfite kit (Diagenode, Liège, Belgium). The DNA input varied between the collaborative exercise and is listed in Table 1 for both Phases. Bisulfite conversions of samples for model evaluation were carried out using 58–100 ng DNA (VISAGE) or 20 ng DNA (INFER). All samples were eluted in 10 µl elution buffer, and 8 µl of bisulfite converted DNA was subjected to the multiplex PCR, targeting a panel of eight age-related

Table 2
CpG sites included in the initial VISAGE age predictive models [31].

Marker	CpG no	Genomic location [GRCh38]	Age model
KLF14	C1	chr1:207823672	Buccal cells
KLF14	C3	chr7:130734372	Bones
KLF14	C4	chr7:130734375	Blood
TRIM59	C8	chr3:160450202	Blood
MIR29B2CHG	C1	chr1:207823681	Blood
MIR29B2CHG	C3	chr1:207823672	Buccal cells
FHL2	C1	chr2:105399282	Blood
ELOVL2	C2	chr6:11044655	Bones
ELOVL2	C7	chr6:11044634	Blood/bones
ELOVL2	C9	chr6:11044628	Buccal cells
EDARADD	C1	chr1:236394382	Buccal cells
PDE4C	C4	chr19:18233105	Bones
PDE4C	C5	chr19: 18233127	Blood/buccal cells
PDE4C	C6	chr19:18233131	Buccal cells
ASPA	C1	chr17:3476273	Bones

Table 1
Allocation of samples across laboratories.

Phase	Validation test	Duplicates	DNA input	Sample type	Samples group A[a]	Samples group B[b]	N per lab	Replicates[c]
Phase 1 (5 laboratories)	Reproducibility/ linearity	No	200 ng	DNAm control	100 %, 75 %, 50 %, 25 %, 10 %, 5 % and 0 %		7	5
	Sensitivity	Yes	50 ng 20 ng 10 ng 1 ng	DNAm control	100 %, 75 %, 50 %, 25 % or 0 %		8 (1 ratio/ lab.)	1
	Concordance	Yes (ex. Lab 1)	200 ng	DNA extract	Blood extract A Saliva extract A	Blood extract B Saliva extract B	2 or 4	3
	Mock casework	Yes (ex. Lab 1)	up to 200 ng	Stain on FTA card	Blood stain A Saliva stain A	Blood stain B Saliva stain B	2 or 4	3
Phase 2 (6 laboratories)	Reproducibility/ linearity	No	200 ng	DNAm control	75 %, 50 % and 25 %		3	6
	Sensitivity	No	20 ng 15 ng 10 ng 7.5 ng 5 ng 1 ng	DNA extract	Blood_7	Blood_2	6	3
	Concordance/ mock casework	Yes	Up to 200 ng	Swab	Blood_1 Blood_2 Blood_3 Blood_4 Blood_5 Saliva_1 Saliva_2 Saliva_3 Saliva_4 Saliva_5	Blood_6 Blood_7 Blood_8 Blood_9 Blood_10 Saliva_3 Saliva_5 Saliva_6 Saliva_7 Saliva_8	20	3

[a] Group A: Phase 1: Labs 1, 2, 4, Phase 2: Labs 4, 5, 6
[b] Group B: Phase 1: Labs 1, 3, 5, Phase 2: Labs 1, 2, 3
[c] Replicates across laboratories: for analyses performed in duplicate, the arithmetic mean of the two replicates was used for inter-laboratory comparisons

markers for somatic cells (Table 2) as described by Wozniak *et al.* (2021) [25].

2.6. Sensitivity

First sensitivity tests (Phase 1) were performed using DNAm control samples. Each laboratory was supplied with one DNAm control sample with an expected DNAm% level of 0 %, 25 %, 50 %, 75 % or 100 %. Bisulfite conversions were performed in duplicates with 50 ng, 20 ng, 10 ng and 1 ng DNA input. In Phase 2, DNA was extracted using the ChargeSwitch® Forensic DNA Purification Kit (TFS) from two blood samples of good DNA quality (Blood_7 for group A and Blood_2 for group B; Table 1**)** to test DNAm quantification and age prediction accuracy with 20 ng, 15 ng, 10 ng, 7.5 ng, 5 ng and 1 ng DNA input into bisulfite conversion. Dilution series were prepared by each laboratory separately, following agreed-upon protocols.

2.7. MPS sequencing

DNAm analysis, including the laboratory workflow for MPS sequencing and the bioinformatic pipeline for data analysis was performed using the VISAGE Enhanced Tool for age estimation from somatic cells, as described in. (2021) [25]. In brief, library preparation using the KAPA Hyper Prep Kit with KAPA Library Amplification Primer Mix was performed with the KAPA SI Adapter Kit Set A+B (Phase 1) or the KAPA Dual-Indexed Adapter Kit (Phase 2) at 15 µM (all Roche, Basel, Switzerland). MPS sequencing on the MiSeq FGx platform was carried out with the MiSeq FGx Reagent Kit (both Verogen, San Diego, CA, USA) using 2 × 200 cycles. Libraries were diluted to 7 pM and spiked with 2 µl 20 pM PhiX control v3 in Phase 1. This was adapted to a 12 pM pool with 30 µl PhiX spike-in in Phase 2.

2.8. Age prediction and data analysis

All FASTQ files obtained from the MiSeq FGx runs were sent to the organizing laboratory for a combined analysis. Alignment, read extraction and calculation of DNAm%-values at model CpG sites (Table 2) were performed as described in [25]. The chronological age of all individuals was estimated using the VISAGE age predictive models for blood or buccal cells [25]. Statistical analysis was performed in Excel and base R [29]. Comparison of linear models was performed using the following packages: "performance" for model performance comparison and "emmeans" for calculations of confidence intervals, estimated marginal means and pairwise contrasts plus correction for multiple comparisons.

2.9. Species specificity

Ten animal DNA samples (dog, cattle, sheep, pig, mouse, rat, goat, horse, cat, orangutan) deriving from an in-house sample collection [30], were bisulfite converted using 100 ng DNA and amplified following the VISAGE Enhanced Tool protocol described in [25] alongside human gDNA as an amplification control. Amplification products were analysed using the Bioanalyzer instrument using a DNA 1000 Kit (both Agilent Technologies, Santa Clara, California, USA) for electrophoretic separation. All primer pairs of the VISAGE Enhanced Tool were run against the *in silico* bisulfite treated reference genomes of the tested species using BiSearch [31]. No *in silico* bisulfite converted genomes were available for orangutan and goat. The obtained sequences of possible in-silico PCR products from BiSearch were compared to the human amplicon sequences using the BLAST [32] program "Needleman-Wunsch alignment of two sequences" to assess homology.

3. Results and discussion

To thoroughly validate the VISAGE Enhanced Tool, a comprehensive inter-laboratory evaluation was conducted, encompassing both the molecular assay and its associated statistical models. The collaborative exercises addressing the assay was structured into two distinct phases, each addressing different aspects of the assay's performance. Phase 1 focused on a developmental evaluation, testing the assay with artificially methylated DNA control samples and further included only two blood and two saliva samples. Phase 2 extended this evaluation by a larger number of blood and saliva samples, to better understand DNAm quantification variation at model CpG sites (Table 2) and their impact on the final age estimation. An overview of the collaborative exercises and the samples assigned to participating laboratories is shown in Table 1. Furthermore, species specificity was addressed at PCR-level and by *in silico* analysis. For evaluation of the age prediction models, samples independent from the original VISAGE testing set [25] were analysed. In total, 160 blood samples and 100 buccal swab samples were sequenced in three laboratories.

3.1. General assay performance

Quality metrics of MiSeq FGx runs were collected for Phase 1 (N = 5) and Phase 2 (N = 6), except for one laboratory, which did not share the files required for an analysis with the Illumina Sequencing Analysis Viewer software. Cluster density varied between 309 and 1153 K/mm2 in Phase 1. Generally, high clusters passing filters were observed, and only one run showed a percentage of clusters passing filter below 90 % (81.19 % ± 14.51), deriving most likely from one tile with no clusters passing filter at all. Aiming at the cluster range recommended by the ForenSeq UAS (400 – 1650 K/mm²), the concentration of the library pool and spiked PhiX were increased in Phase 2–12 pM and 5 %, respectively. No drops in cluster density below 400 K/mm² were observed for recorded quality metrics in Phase 2, while high percentages of cluster passing filter (> 90 %) and reads with a Q30-score > 80 % were maintained for all runs of Phase 2. Quality parameters of all runs are listed in Supplementary Table S2**.**

An initial assessment of the read depth was performed based on the seven DNAm control samples analysed by five laboratories in Phase 1 (N = 35 samples). Because neighbouring model CpG sites located on the same amplicon are fully covered during paired end sequencing by both directions, only minor differences in read depth are expected and the average read depth was used for analysis. As expected, the mean read depth at target CpG sites correlated with the cluster density of the respective run and varied from 21,104.22 paired reads (cluster density = 309 K/mm2) to 102,965.12 paired reads (cluster density = 1144 K/mm2). Due to high read depth outliers, the median was used for a general comparison of read depth per amplicon. The lowest median read depth was obtained for *PDE4C* with 19,665 paired reads, the highest for *FHL2* with 71,153 paired reads. The previously set threshold of 1000 reads for the VISAGE age tools was reached for 99.62 % of all amplicons, with only one reproducibility sample dropping below 1000 reads at the *MIR29B2CHG* amplicon (678 paired reads). The respective sample was included in the run showing the lowest cluster density. Analysis of the read distribution among the eight amplicons was performed by normalizing the read depth of one amplicon by the total amount of reads per sample (Fig. 1**A**). Overall, *PDE4C*, *MIR29B2CHG*, *EDARADD* and *ASPA* showed a lower number of reads compared to *KLF14*, *ELOVL2*, *FHL2* and *TRIM59*.

Additionally, concordance samples from Phase 2 (N = 120) were used to analyse read depth and distribution with blood and saliva samples. Surprisingly, the normalized read depth showed a different picture compared to DNAm control samples, with the most reads attributed to the FHL2 amplicon, and a more even distribution for the remaining amplicons (Fig. 1**B**). To determine whether these differences are assay-related or sample-related, the normalized read depth for DNAm controls of Phase 2 (N = 18) was calculated as well. As for Phase-1 DNAm control samples, the read distribution showed a lower number of reads at *PDE4C*, *MIR29B2C*, *EDARADD* and *ASPA*, but also for

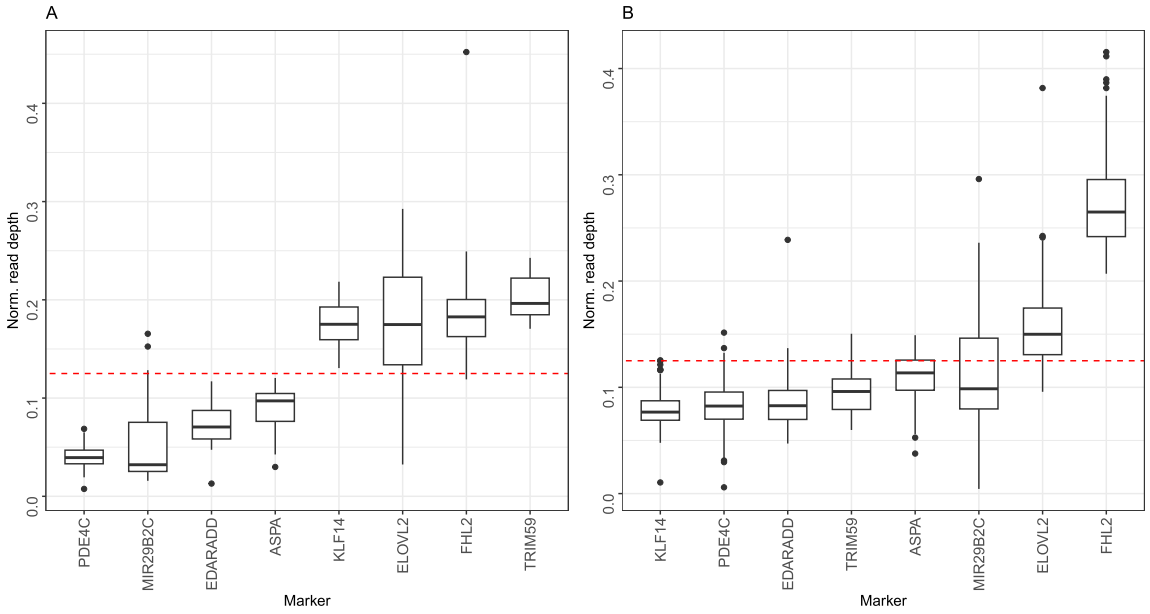


Fig. 1. Normalized read depth for **(A)** Phase 1 reproducibility samples (N = 35) and **(B)** Phase 2 concordance samples (N = 120). The red dotted line indicates the expected normalized read depth if all amplicons were equally represented (0.125). Outliers are defined as points beyond 1.5 × IQR (interquartile range) from the quartiles.

TRIM59. The lower number of reads at *TRIM59* might be attributed to a technical change during Phase 1 and 2 (e.g. new primer mixes). However, except for *TRIM59* all controls showed a similar read distribution with a lower amount of reads at *PDE4C* and *MIR29B2C*. Furthermore, differences between DNAm control samples and DNA extracts from semen were also observed for the VISAGE enhanced tool for age estimation from semen samples [33], indicating an effect of sample type on primer and sequencing efficiency.

Furthermore, base misincorporation rates (percentage of As and Gs at target Cs), and bisulfite conversion efficiency (average T read percentage at non-CpG Cs) were used to assess sequence quality and reliability of the obtained DNAm-values. With a mean percentage of base misincorporation of 0.04 ± 0.05 for DNAm controls and 0.06 ± 0.1 for concordance samples and a bisulfite conversion efficiency above 99.11 % for all samples, both parameters were unlikely to affect the final DNAm quantification.

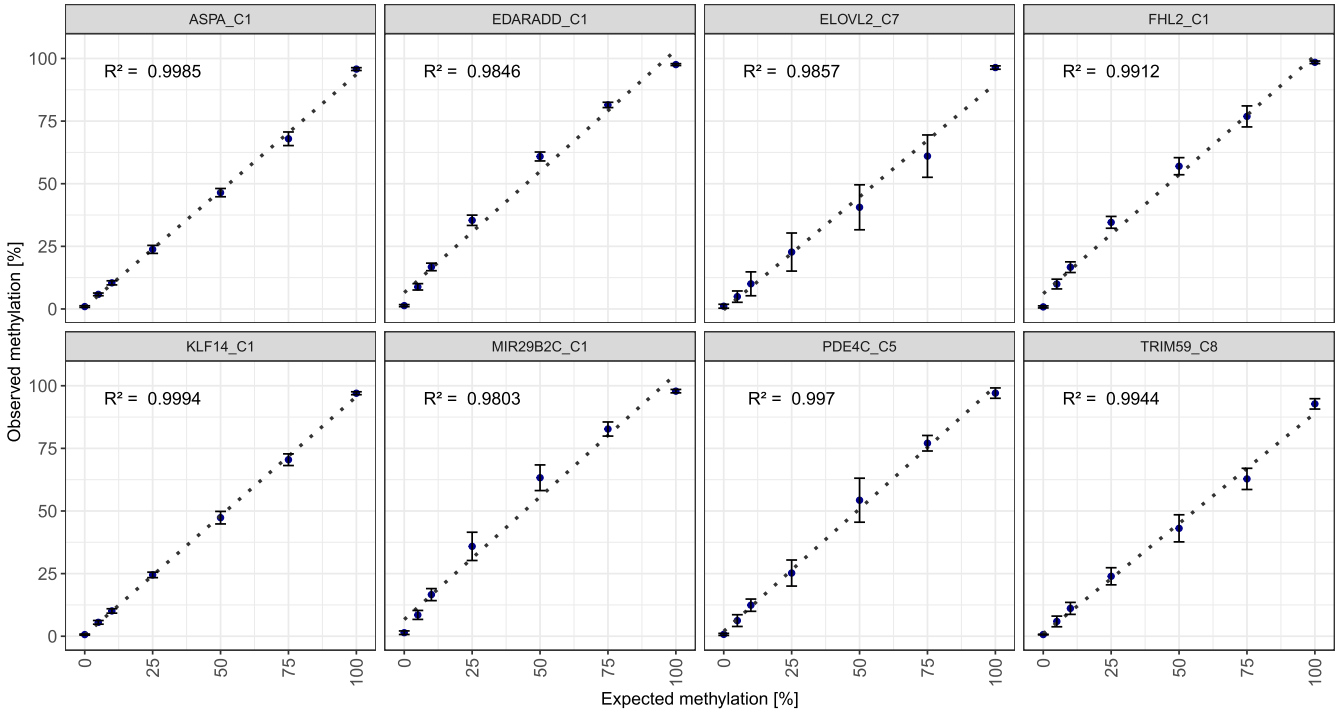


Fig. 2. Phase 1. Measured DNAm levels of artificially methylated control samples plotted against the expected DNAm. Data for a single CpG site per marker are shown. Data of all model CpG sites is illustrated in Supplementary Fig. 1.

3.2. Reproducibility and linearity tests

Seven DNAm standards were prepared to evaluate reproducibility and linearity of the VISAGE Enhanced Tool. Within DNAm standards, CpG sites located on the same amplicon (Table 2) are expected to show the same methylation levels. Comparison of these CpG sites showed only small differences for *ELVOL2* and *PDE4C* with an average SD of 0.73 and 1.41 DNAm% (3 target CpGs/amplicon). Neighbouring CpG sites within the KLF14-amplicon showed a higher variation with an average SD of 2.55 DNAm% (3 target CpGs/amplicon). For *MIR29B2CHG* (2 target CpGs/amplicon) a mean difference of only 0.42 DNAm% was observed.

Obtained DNAm levels of all controls were concordant among laboratories (N = 5; Fig. 2). The highest variation among laboratories was found at an expected DNAm% level of 50 % with a mean SD across model CpG sites of 4.79, followed by 25 % and 75 % (mean SD = 3.58 and 3.53), 10 % (mean SD = 2.14), 5 % (mean SD =1.69), 100 % (mean SD = 0.93) and 0 % (mean SD = 0.42). Mean standard deviations across controls ranged from 1.17 DNAm% for *ASPA*_C1 to an average of 4.74 DNAm% for CpG sites located on the *ELOVL2* amplicon. Notably, these amplicons represent the shortest (*ASPA* = 108 bp) and the longest (*ELOVL2* = 267 bp) amplicons within the PCR multiplex.

When comparing the expected and measured DNAm%-values, high coefficients of determination were observed for all age markers ($R^2 \geq 0.98$; Fig. 2). Overall, DNAm controls showed a good agreement between the two values and only slight amplification biases towards an over- (e.g. *MIR29B2C*), or underestimation (e.g. *ELOVL2*) of the expected DNAm level were visible. These results indicate a stable performance of the assay over the whole range of DNAm ratios and confirm the observations made during assay design [25].

Furthermore, three DNAm controls (DNAm% = 25, 50 and 75 %) were included in each run of Phase 2, to continuously monitor the assay for robust DNAm quantification. Results were highly comparable to Phase 1, with average SDs ranging from 1.65 (*ASPA*) to 4.12 DNAm% (*ELOVL2*) and high coefficients of determination ($R^2 > 0.98$; Supplementary Fig. S2).

3.3. Technical variability

Technical variability was assessed by calculating the absolute difference in DNAm%-values between duplicates in the concordance and mock casework studies. Here, the term 'duplicates' applies to all sample-replicates that were processed in parallel following DNA extraction, as the bisulfite conversion is known to cause DNA loss and degradation [34]. Differences in DNAm%-values were calculated for the respective model CpG sites of blood (N = 6 CpGs) or saliva (N = 5 CpGs) samples. Across markers and laboratories, the mean difference between measured DNAm%-values of duplicates was 0.94 ± 1.64 DNAm% for blood samples and 1.26 ± 2.4 DNAm% for saliva samples (N = 38). Mean differences between duplicates were below 1.8 % for all model CpG sites. The absolute differences between obtained DNAm%-values are plotted in Supplementary Fig. S3.

High differences were observed only for two stain samples of Phase 1, with one blood stain and one saliva stain showing high differences (mean diff. = 6.17 and 4.32 DNAm%, respectively). When looking into the laboratory workflow, quantification results after the multiplex PCR were lower for the saliva stain duplicates (~6 ng/µl vs. 28 ng/µl for DNA extracts), which might explain the high differences. No deviations during the laboratory workflow or from sequencing quality parameters were noticed for the blood stain. With a maximum difference of up to 19.42 DNAm% and 21.98 DNAm%, results from both stains are likely to show large deviations in the final age prediction outcome. Although the overall DNAm quantification was robust, the two casework stain samples demonstrate the importance of performing technical replicates for DNAm quantification. Since MPS tools are technically capable of generating high-quality sequence reads from samples with insufficient DNA quantity or poor DNA quality for age prediction [35], unreliable quantifications may go undetected in individual reactions.

3.4. Concordance and mock casework samples

During Phase 1, DNA extracts and stains from the same two blood and saliva samples were sent to participating laboratories to assess whether the DNA extraction step leads to increased variability. As a first quality control step, the difference between the age predictions was calculated for samples performed in duplicates. Notably, no duplicates were performed by Lab 1, to ensure for an equal number of samples per flow cell (Table 1). Observed differences between age estimates reflected the technical variability in DNAm quantification. A low mean difference was obtained for DNA extracts (1.38 ± 1.14 years, N = 8), which increased for stain samples to 4.40 ± 8.88 years difference (N = 8). This was mainly caused by one blood stain (Sample B), which already showed large differences between DNAm quantifications. With a difference of 26.04 years between duplicates, the sample was considered to have failed quality control and was excluded from further analysis. Mean age prediction results were compared among laboratories (Table 3) and showed highly concordant results for DNA extracts ("Concordance", mean SD = 1.36 years) and as stains on FTA cards ("Mock Casework", mean = 2.55 years). Additionally, age predictions obtained from DNA extracts and stains of the same sample were close to each other (mean diff. = 2.04 ± 1.94 years).

To further investigate mock casework samples, swabs from ten blood and eight saliva samples were analysed in the second evaluation phase. The laboratories were divided into two groups of three (A & B), which tested different samples (Table 1). Each group received five blood and five saliva samples. Differences between age estimates of duplicates were low for blood (0.82 ± 0.62 years) as well as saliva samples (0.88 ± 0.62 years). Age estimation of blood (Fig. 3) showed only low differences among laboratories (mean SD = 0.63 years) with a maximum difference of 1.20 years between age predictions obtained from laboratories. Overall, blood estimates showed a MAE of 4.92 years.

Because availability of saliva samples was restricted (N = 8), two samples were sent to two laboratories of both groups resulting in six instead of three replicates (Fig. 3). Although age prediction results among laboratories were also highly concordant with a low mean SD of 0.80 years and a maximum difference between laboratories of 1.57 years, a constant overestimation of the chronological age was observed (mean error and MAE = 8.04 years). Saliva samples were selected as casework type samples as buccal swabs do not represent a realistic trace found at a crime scene. However, as no VISAGE age estimation model is currently available for saliva, age predictions were made with the predictive model for buccal swabs. Previous studies have described differences in the cell composition between the two sample types, as well as in the strength of the correlation of DNAm with chronological age [7,36]. This might explain the overestimation of the chronological age and indicates the need for a separate predictive model for saliva samples.

Table 3
Mean predicted age (± SD; N = 3) of concordance and mock casework samples (Phase 1).

Sample Type	Sample	Chron. Age [years]	Concordance [years]	Mock casework [years]
Blood	Sample A	20	20.23 ± 1.44	19.85 ± 0.82
Blood	Sample B	49	51.27 ± 1.92	52.23 ± 2.55[a]
Saliva	Sample A	52	66.71 ± 1.79	70.09 ± 3.94
Saliva	Sample B	27	25.87 ± 0.29	24.94 ± 2.87

[a] one outlier was excluded, N = 2

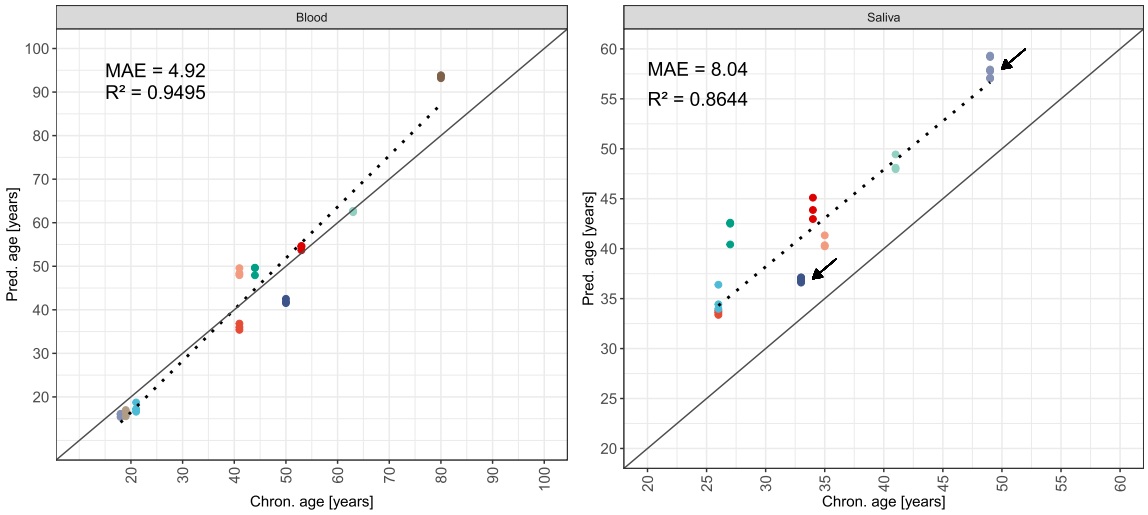


Fig. 3. Phase 2. Comparison of estimated and chronological age for blood and saliva concordance samples. The dotted line represents the best fit line, while the diagonal solid line indicates perfect match between predicted and actual age. Two saliva samples, which were sent to all six laboratories, are highlighted with arrows. Concordance samples are distinguished by different colors (for color interpretation in this figure, please refer to the web version of the article).

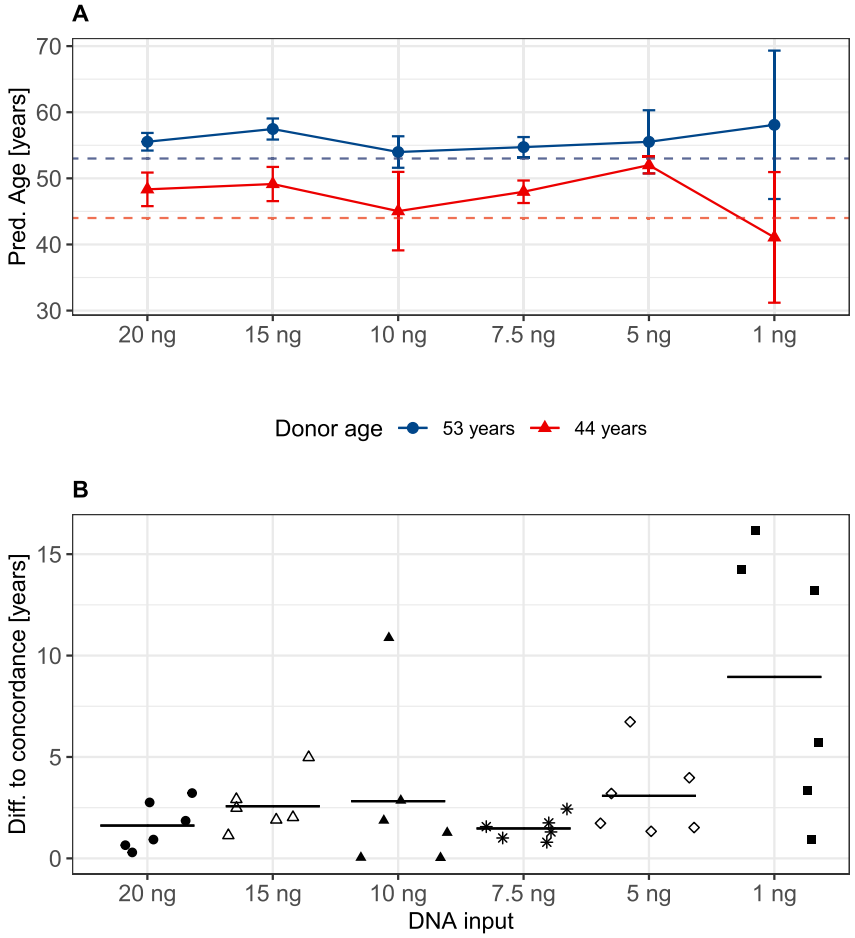


Fig. 4. Phase 2. (**A**) Mean estimated age of dilution series of two good quality blood samples from a 44- and a 53-year-old sample donor (N = 3, per DNA input). Error bars represent the standard deviation, the chronological age is indicated by horizontal dashed lines. Note that no measurements were taken between the points. (**B**) Difference between age estimations of the sensitivity samples (N = 6 per DNA input) and reference samples. The mean difference for each DNA input is indicated by a horizontal line.

3.5. Assay sensitivity

Assay optimization and initial evaluations of the VISAGE age tools were performed with 200 ng DNA for bisulfite conversion, based on the DNA input recommended by the manufacturer. To test the assay with forensically relevant DNA amounts, dilution series down to 1 ng DNA input into bisulfite conversion were tested. DNA loss and degradation frequently occur during bisulfite conversion, resulting in variable DNA yields and up to ~50 % DNA loss in the final eluate [34,37]. To assess sensitivity of the VISAGE Enhanced Tool, studies were performed with DNAm control samples (0 %, 25 %, 50 %, 75 %, and 100 %) to test for linearity (Phase 1) and with two high quality blood samples to assess for quantification and age prediction accuracy with lower DNA inputs (Phase 2).

DNAm control samples were run at 50 ng, 20 ng, 10 ng and 1 ng DNA input and analysed in duplicates. Two samples of the 50 %-methylated controls (20 ng and 10 ng), had to be excluded due tosample loss during library preparation, leaving only one replicate available for analysis. Read depth was sufficient at all CpG sites except for eight amplicons at 1 ng DNA input with a read depth < 1000 paired reads. This affected mainly CpGs located at *PDE4C*, for which the read depth was reduced at 1 ng (Supplementary Fig. S4). The average difference between measured DNAm%-levels of duplicates was similar at 50 ng and 20 ng (mean = 2.33 and 2.76, respectively) and increased at 10 ng to 4.39 DNAm%. At 1 ng, quantification accuracy was already impaired and showed large differences between the technical replicates (mean = 12.28 DNAm%). For a comparison with reproducibility samples (200 ng DNA input; see 3.2), the obtained mean DNAm-values were taken as a reference (N = 5) and compared with mean DNAm% results from sensitivity samples (N = 2). Good agreement between the reference, and low DNA input samples was observed for most markers down to 10 ng, whereas at 1 ng DNAm quantifications showed severe outliers (Supplementary Fig. S5). The mean absolute difference in DNAm% to the reference across markers was 2.2 ± 2.8 at 50 ng, 2.7 ± 4.4 at 20 ng, 3.4 ± 4.4 at 10 ng and 9.1 ± 11.3 at 1 ng DNA input. Results are plotted for each marker in Supplementary Fig. S6.

Dilution series of two concordance blood samples (44- and 53-year-old) focused on a DNA input range between 20 and 1 ng (Table 1). Both samples were analysed by three laboratories (N = 3, per sample and DNA input) and compared to mean measured DNAm%-values and age estimates obtained during the concordance study (Fig. 4). The average absolute difference from DNAm quantifications to the reference across markers and both samples was 1.89 DNAm% at 20 ng DNA input and increased by 0.95 DNAm% throughout the dilution series to 5 ng (mean = 2.84 DNAm%). In contrast, at 1 ng, high differences were observed (mean diff. = 7.91 DNAm%). Similarly, the variability of DNAm quantifications among the three laboratories was low down to 5 ng DNA input (mean SD = 3.18 DNAm%), whereas increased variability was observed for 1 ng DNA input (mean SD = 8.13 DNAm%). In terms of age prediction accuracy, the observations were analogous to results from DNAm quantification. With the exception of one outlier within the 10 ng samples of the 44-year-old participant, neither the standard deviation nor the mean difference from the concordance sample increased substantially from 20 to 5 ng DNA input (Fig. 4). In contrast, age estimates from 1 ng samples showed high differences among laboratories as well as to the reference. Notably, the mean read depth ranged from 42,409.1 reads at 10 ng to 61,393.4 reads at 5 ng and no overall decrease in read depth for lower DNA inputs was observed. These results suggest that the age prediction accuracy can be maintained down to 5 ng DNA input into bisulfite conversion, but good quality DNA samples and replicates are required to ensure reliable results. Although the sequencing performance did not decrease at 1 ng for blood samples, the obtained DNAm quantifications were distorted and not representative for the initial DNAm level of the blood sample.

3.6. Species specificity

Amplification products were detected in the ten bisulfite converted animal DNA samples, when examining Bioanalyzer electropherograms. However, none of the obtained electropherograms matched the profile expected for human DNA (**Supplementary File 1**). The observed amplification products could either be due to non-specific primer binding, facilitated due to the low complexity of bisulfite-converted DNA, dimer formations, high abundance of A and T nucleotides [31], or conserved genomic regions. To gain further insights into unspecific amplification, an *in-silico* analysis using BiSearch and BLAST was conducted. An overview of possible PCR products from animal DNA and the comparison with the human amplicon sequences is shown in Table 4. BiSearch PCR products from *ASPA* and *EDARADD*, when queried against rat and cat genomes, exhibited low sequence identity compared to the respective human genome amplicon sequence. Additionally, primers showed a high number of mismatches (up to four), which makes it unlikely to yield enough PCR products and align successfully against the human reference to influence final DNAm levels. In contrast, BiSearch results for *MIR29B2CHG* and *ASPA* produced amplicon sequences with a high level of sequence identity (>91 %) for sheep, dog and cat (*MIR29B2CHG*) and cattle, dog, horse, pig and sheep (*ASPA*). When inspecting the target CpG site of the sequences, the G of the CpG site of *ASPA* is not homolog (A instead of G) in cattle, dog, pig and sheep thus preventing DNAm quantification. BiSearch PCR products for *MIR29B2CHG* hit the antisense strand for sheep and cat, which would be noticeable when analysed. From *in silico* analysis, an unspecific amplification of *ASPA* and *MIR29B2CHG* of dog and horse DNA leading to mapped sequence reads cannot be excluded. Thus, a careful inspection of alignments (e.g. using IGV) should be performed, particularly if animal-human mixtures are expected.

3.7. Evaluation of the VISAGE Enhanced Tool age statistical models for blood and buccal swabs

Two sample sets that were not used for predictive model training and testing at the Jagiellonian University in Poland [25], were available for evaluation of the age models for blood and buccal swabs. The first sample set (VISAGE-evaluation; 49 blood samples, 47 buccal swabs) was sequenced in single reactions by Lab 1. The second sample set was collected for the INFER project, which focused on the implementation of the VISAGE tools in governmental laboratories and comprised 111 blood samples and 53 buccal swabs. Samples of the INFER sample set were analyzed in duplicates by three laboratories (Lab 1, 4 and 7) and mean age prediction results were used for analysis. The mean difference between duplicates was 2.4 ± 1.8 years for blood and 2.6 ± 2.0 years for buccal cell samples. Additionally, the INFER sample set contained three samples (18, 44 and 64 years-old) which were sequenced by all three laboratories to check for a stable assay performance and the mean calculated age estimates were used (SD = 1.6 years for blood and 1 year for buccal swab samples, Supplementary Table S3**).** For age model evaluation both sample sets were combined, resulting in 160 blood samples (49 % female) and 100 buccal swab samples (43 % female; Supplementary Table S4).

The VISAGE Enhanced Tool models estimated the chronological age with a MAE of 3.95 years for blood (RMSE = 5.0 years) and 4.42 years for buccal swabs (RMSE = 5.59 years) using the respective prediction model (Fig. 5 and Table 5). Comparing the obtained MAEs to the MAEs of the original VISAGE testing set (MAE = 3.2 and 3.7 years; respectively N = 48 [25]), an increase of approximately 0.7 years (8.4 month) was noted. Overall, the blood predictive model showed a tendency to underestimate the chronological age (mean error = −1.8 years), whereas the predictive model for buccal cells on average predicted a higher age (mean error = 2.3 years). When comparing the percentage of samples classified as a correct age prediction within pre-defined error margins (Table 6), 88 % of blood samples and 81 % of buccal swabs showed an

Table 4
In silico primer analysis results using BiSearch and BLAST.

Primer pair	Species	Target DNA strand	Amplicon size	Target DNA strand (human)	Amplicon size (human)	Percent identity	Target CpG homolog
MIR29B2CHG	Sheep	-	145	+	146	96 %	yes
	Dog	+	146			96 %	yes
	Cat	-	147			96 %	yes
EDARADD	Cat	+	137	-	193	29 %	no
ASPA	Cattle	+	108	+	108	94 %	no
	Dog	+	108			94 %	no
	Horse	+	108			91 %	yes
	Pig	+	108			93 %	no
	Sheep	+	108			94 %	no
	Rat	-	290			61 %	no

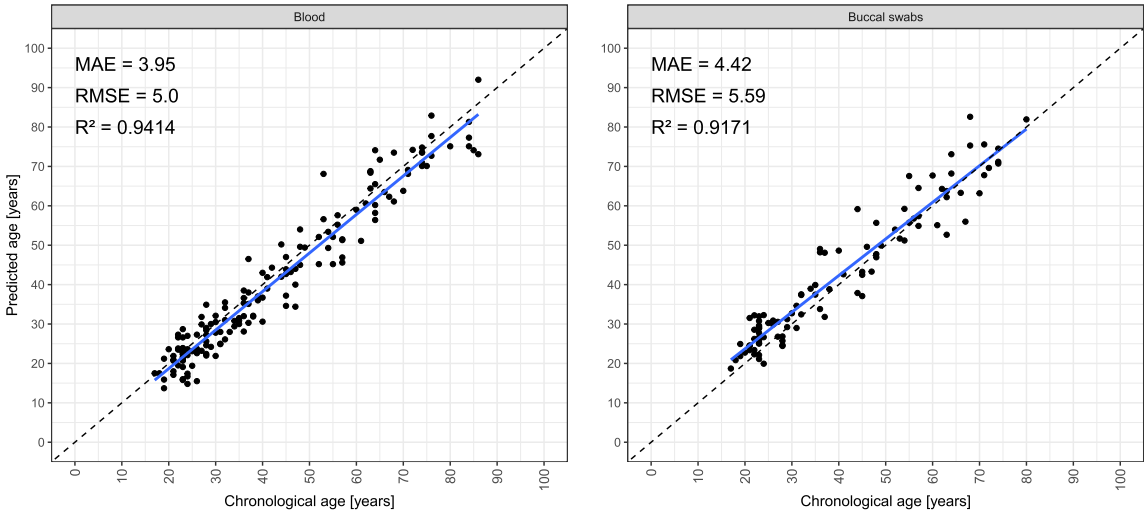


Fig. 5. Age estimation results of DNA extracts from blood (N = 160) and buccal swabs (N = 100). The solid blue line represents the linear trend, while the dashed line denotes the line of identity. MAE = mean absolute error, RMSE = root mean squared error.

Table 5
Age estimation results.

Age category[a]	Age range	Blood			Buccal Swabs		
		N	error	abs. error	N	error	abs. error
1	< 20	6	−0.42	2.52	5	3.21	3.21
2	21–40	86	−1.74	3.41	53	3.45	4.46
3	41–60	33	−2.22	4.63	23	1.58	4.15
4	> 60	35	−2.43	3.79	19	−0.22	4.98
overall	17–86	160	−1.80	3.95	100	2.31	4.42

[a] Age ranges were defined according to Wozniak et al. (2021).

Table 6
Proportion of age predictions with absolute error below pre-defined margins.

Sample type	% < 2.5 years	% < 5 years	% < 7.5 years	% < 10 years	% < 12.5 years	% < 15 years[a]
Blood	40.00	67.50	88.10	94.40	98.10	99.40
Buccal Swabs	35.00	66.00	81.00	90.00	96.00	99.00

[a] The maximum absolute error was 15.2 years for both sample types.

error of less than 7.5 years. No significant difference was found between age estimations for females and males in blood (MAE females = 4.06 years and MAE males = 3.85 years, Wilcoxon rank-sum test: p-value = 1) and buccal swab samples (MAE females = 3.92, males = 4.80; Wilcoxon rank-sum test: p-value = 0.42).

Because age estimation is typically less accurate in older individuals [6,14,38,39], samples were categorized in four groups (Table 5). Age ranges of the four categories were selected based on the VISAGE testing set [25], to allow for a comparison between studies. In contrast to samples used for predictive modelling, no children were available in this study and no conclusions can be drawn for age category 1 (0–20-years-old). Also, it should be noted that a higher number of individuals was available in age category 2 (21–40-years-old). Surprisingly, for this blood sample set, the highest MAE was obtained for 41–60-year-olds (MAE = 4.6 years) and not for > 60-year-old individuals (MAE = 3.8 years). Buccal swabs showed a stable prediction error from 21 to 60-year-old individuals, which increased as expected for > 60-year old individuals.

3.7.1. Inter-individual differences

Blood and buccal swab samples were collected from each donor (N = 100), and age estimation results from the same donor were compared between the two sample types. To explore whether age estimation errors might derive from a higher or lower epigenetic age compared to the chronological, the distribution of the errors was compared between paired samples. However, no clear conclusions could be drawn from this comparison (Supplementary Fig. S7). To explore further, the top 10 % of samples (N = 10) with the highest absolute age estimation error were selected. The MAE for these 10 samples was 7.66 years for blood samples (mean chronological age = 49.6 ± 14.9 years) and 12.07 years for buccal swabs (mean chronological age = 44.9 ± 17.54 years). Only two samples were assigned to the high-error group for both tissue types. When looking at the prediction error in the respective other tissue, the MAE for blood was the same as it was for the overall sample set (MAE = 3.95 years) and the MAE for buccal swabs

was elevated by 0.9 years (MAE = 5.34 years). When further exploring the direction of the error, for half of the samples, the age estimation error exhibited opposite directions in the two tissue types. This suggests that even if the high age estimation error was linked to a higher or lower epigenetic age, it appears not to be present in both tissue types.

3.7.2. Inter-laboratory differences

Although all laboratories used the same commercially available kits and received a detailed protocol for DNAm analysis, small differences in the workflow might influence the DNAm measurements and consequently the final age prediction outcome. Therefore, age prediction results were compared among laboratories (Supplementary Figure S8). For blood samples, a substantially higher MAE was calculated for Lab 4 (MAE = 5.01 years; N = 71) compared to the two other laboratories (Lab 1: MAE = 2.85, N = 64 and Lab 2: MAE =3.69 years, N = 22). For buccal swabs, the highest MAE was also calculated for Lab 4 (MAE 5.75 years, N = 15), followed by Lab 1 (MAE = 4.36 years; N = 61) and Lab 7 (MAE = 2.93 years, N = 21). Differences in the MAE might be explained by the different number of individuals and the different distribution of the chronological age in each laboratory. However, whereas the mean error, was almost the same for all three laboratories for buccal swabs (~2.3 years), Lab 4 showed not only a higher MAE for blood samples, but also a stronger underestimation of the chronological age (mean error = −3.64 years) compared to the other laboratories (mean error =-0.01 and −1.77 years).

To further explore, if inter-laboratory differences might influence the final age prediction outcome, age estimations were compared to the original VISAGE testing set [25]. First, a separate linear model was calculated for each laboratory and tissue type, all showing a high correlation between the predicted and the chronological age (Supplementary Table S5). Next, a unified linear model was computed, without considering the laboratory information. This model was then compared to its extended version that included the laboratory information as an additional explanatory variable. While the inclusion of the laboratory information showed no significant difference in model performance for buccal swab samples (p-value = 0.18; ANOVA), accounting for the laboratory information significantly improved the model performance on the blood data (p-value < 0.001; ANOVA). *Post-hoc* analysis of the laboratory effect in blood samples revealed that the difference compared to the VISAGE model testing set was significant only for Lab 4 (p-value < 0.001; Table 7), corroborating the observation that this laboratory had the highest mean underestimation of age among all participating laboratories.

Additionally, estimated marginal means for chronological ages from 0 to 90-years were calculated in 5-year increments. Unlike Lab 1 and Lab 7, the 95 % confidence interval (CI) for the estimated means of Lab 4 did not include the chronological age (Supplementary Table S6), further indicating the presence of a bias in Lab 4. Moreover, pairwise contrasts of the laboratories calculated for the same chronological age intervals showed a significant difference between the original VISAGE testing set and Lab 4, as well as between Lab 4 and Lab 1 (Supplementary Table S7).

In summary, while the VISAGE predictive model for buccal swabs was applied to samples sequenced in all laboratories without a significant difference compared to the VISAGE testing set, variability was observed among laboratories for the blood predictive model. Here, Lab 4 showed a clear tendency to underestimate the chronological age with a significant difference to the original VISAGE testing set. These results indicate a bias within the DNAm quantification of Lab 4, although the collaborative exercises performed for the technical evaluation of the VISAGE Enhanced showed concordant results for all laboratories. When excluding Lab 4 from the analysis, the MAE for blood samples decreases to 3.10 years (N = 89), demonstrating a model performance comparable to that observed in the VISAGE testing set.

Table 7
Coefficients of the multiple linear model considering the laboratory information.

Parameter	Beta	Std. Error	Std. Beta	p-value	Lower 95 % CI	Upper 95 % CI
(Intercept)	1.45	0.88		0.0980	−0.27	3.18
Chron. age	0.97	0.02	0.99	< 0.0001	0.94	1.01
Lab 1	−0.57	0.82	−0.01	0.4910	−2.19	1.05
Lab 4	−3.93	0.81	−0.10	< 0.0001	−5.52	−2.34
Lab 7	−1.99	1.11	−0.03	0.0760	−4.18	0.20

Note: The VISAGE original testing set served as reference; Std. Error: standard error, Std. Beta: standardized beta, CI: confidence interval

4. Conclusions

The VISAGE Enhanced Tool successfully performed DNAm quantification in seven laboratories that participated in several collaborative exercises. All laboratories were asked to strictly follow the provided protocol with indicated kits for bisulfite conversion, multiplex PCR, library preparation and sequencing to avoid that differences in the laboratory workflow affect the final DNAm quantification as reported earlier [21–23]. The assay accurately measured the DNAm levels of DNA standards with only little amplification bias for most markers. DNAm quantification showed small differences between technical duplicates of the analysed blood and saliva samples and age prediction results were highly concordant among participating laboratories. High age prediction accuracy was maintained down to 5 ng with good quality DNA extracts from blood samples. The use of technical replicates for DNAm quantification to obtain reliable results is strongly recommended. Moreover, the difference among age predictions from replicates should be reported alongside the mean estimated age. This applies in particular to low DNA input samples, as DNA loss and degradation during the bisulfite conversion is variable [34].

The VISAGE Enhanced Tool age model for buccal swabs overestimated the chronological age in all eight saliva samples. Although only a small sample set was analysed, this might indicate a bias within the model, which most likely derives from the different cell composition between saliva and buccal swab samples [7,36]. As saliva samples represent a relevant sample type for casework applications, larger saliva sample sets should be analysed to adjust the model if required.

When predicting the chronological age of donors based on 160 blood samples and 100 buccal swab samples, the obtained MAEs (blood: 3.95, buccal swabs: 4.41 years) were ~0.7 years higher than the MAEs obtained with the initial VISAGE testing set [25]. However, a significant discrepancy was observed in the performance for blood samples sequenced in Lab 4 compared to the VISAGE testing set. When removing samples analysed at this laboratory, the MAE for blood decreases to 3.10 years. This suggests that in addition to a technical validation of the assay, an in house-validation of the predictive model with reference samples should be performed when implementing the VISAGE Enhanced Tool. No significant differences in performance compared to the VISAGE testing set were noted for the remaining laboratories.

Ultimately, although the presented evaluation generally showed a good assay performance in all laboratories, several challenges for forensic age estimation remain to be overcome. Larger sample sets are required to get more detailed insights into the abundance and nature of biological outliers. Furthermore, DNAm patterns can be influenced by several parameters, like a person's health state, lifestyle choices or biogeographic ancestry [40]. Some of these factors like heavy alcohol consumption [26], Alzheimer's Disease, Graves' disease [41], chronic lymphocytic leukemia [42] and physical exercise [43] have already been tested for their influence on DNAm patterns of the VISAGE age markers. While some factors showed no effect on the age estimation accuracy (alcohol consumption [26], Graves' disease, late onset Alzheimer [41]), age prediction became less accurate in patients with early onset Alzheimer's Disease [41] and power athletes [43], and was unreliable for individuals with chronic lymphocytic leukemia [42]. However, the impact of many other factors is not yet fully understood and will require the joint effort of laboratories performing DNAm-based age

prediction gain insights into the effect on the final age estimation.

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Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used OpenAI *ChatGPT*, 2024, (https://chat.openai.com) in order to improve language. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.fsigen.2025.103316.

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